

Module 2 Modelling Risk



Modelling Risk

Last week (Module 1), you learned how to measure financial performance over time by calculating and visualising returns. This week, we extend that foundation by turning to a closely related but distinct concept: risk.

In finance, risk is not simply the possibility of loss. More broadly, risk refers to uncertainty about future financial outcomes. When we make financial decisions, we do not know exactly what will happen. Instead, we face a range of possible outcomes, some better and some worse than expected. Two key ideas will guide everything we do this week:

- Financial outcomes are uncertain
- That uncertainty matters for decision-making

Module 2 is designed to build that toolkit in R so you can move from “I can compute returns” to “I can analyse uncertainty, volatility, and downside risk.”

WATCH

[Module 2 Overview Video \[01:43\]](#)



What will you be learning this week?

Key Concepts	Content
2.1 Understanding Risk in Finance	Distinguish risk from return, and see why uncertainty matters real decisions for investors, firms, and institutions.
2.2 Time series properties for risk measurement	Identify stationary vs non-stationary behaviour, recognise random walks, and apply these concepts in the context of prices and returns.
2.3 Volatility as the core measure of risk	Define volatility formally, interpret it as dispersion in returns, and compare risk profiles across assets
2.4 Volatility clustering and	Detect clustering visually, compute realised volatility, use squared

Key Concepts	Content
persistence	returns as a proxy, and quantify persistence using autocorrelation (ACF).
2.5 Downside risk: Value at Risk and Expected Shortfall	Compute historical VaR and ES, interpret both in return and dollar terms, and understand why ES captures tail severity that VaR misses.

How Module 2 supports your Course Learning Outcomes

This module develops your ability to measure, interpret, and communicate financial risk using time series methods in R, supporting:

- **CLO 1:** Explain key concepts and theoretical foundations of financial analytics and their role in strategic decision-making.
- **CLO 2:** Apply quantitative and time-series techniques to model risk, return, and financial performance in real-world contexts.
- **CLO 3:** Use business analytics tools to develop, evaluate, and communicate predictive insights that support financial problem-solving.
- **CLO 4:** Create and communicate financial insights using effective visual, written, and oral formats suited to both technical and non-technical stakeholders.

How much time can you expect to spend on Module 2?

This workload is designed to be manageable and flexible (i.e. you don't need to complete everything in one sitting!).

Estimated learning hours

You will need to spend around 25 hours in total engaging with the learning materials, attending live sessions, studying, and working towards your assessment.